

Cloud Computing Technologies

DAT515 Fall 2026

DevOps: Best Practices and Tools

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DO
DO NOT
PRESS

CI/CD



CODE/CD
SERVERS
CONTAINERS

DO NOT
PRESS

AUTOMATION

STOP

DEPLOYE

DEVOPS

DEVOPS

OUTERATION
AUTOMATION

DEPLOY
DEPLOY

DEPLOYE

DEPLOY
DEPLOY

CUTALTY
PRESS

DEPLOY
PRESS

DO NOT
PRESS

AUTOMATION



What is DevOps?

An organizational culture that
integrates development and
operations for faster, more
reliable software delivery

DevOps is an Organizational Culture

- Break down silos between **development and operations teams** → streamlined software delivery
- **Automation** of workflows → faster, more reliable processes
- **Continuous integration and delivery** → rapid, frequent releases
- Improve software quality through **continuous monitoring** and **feedback**
- **Impact:**
 - Faster deployment of new features
 - Reduced time to fix problems

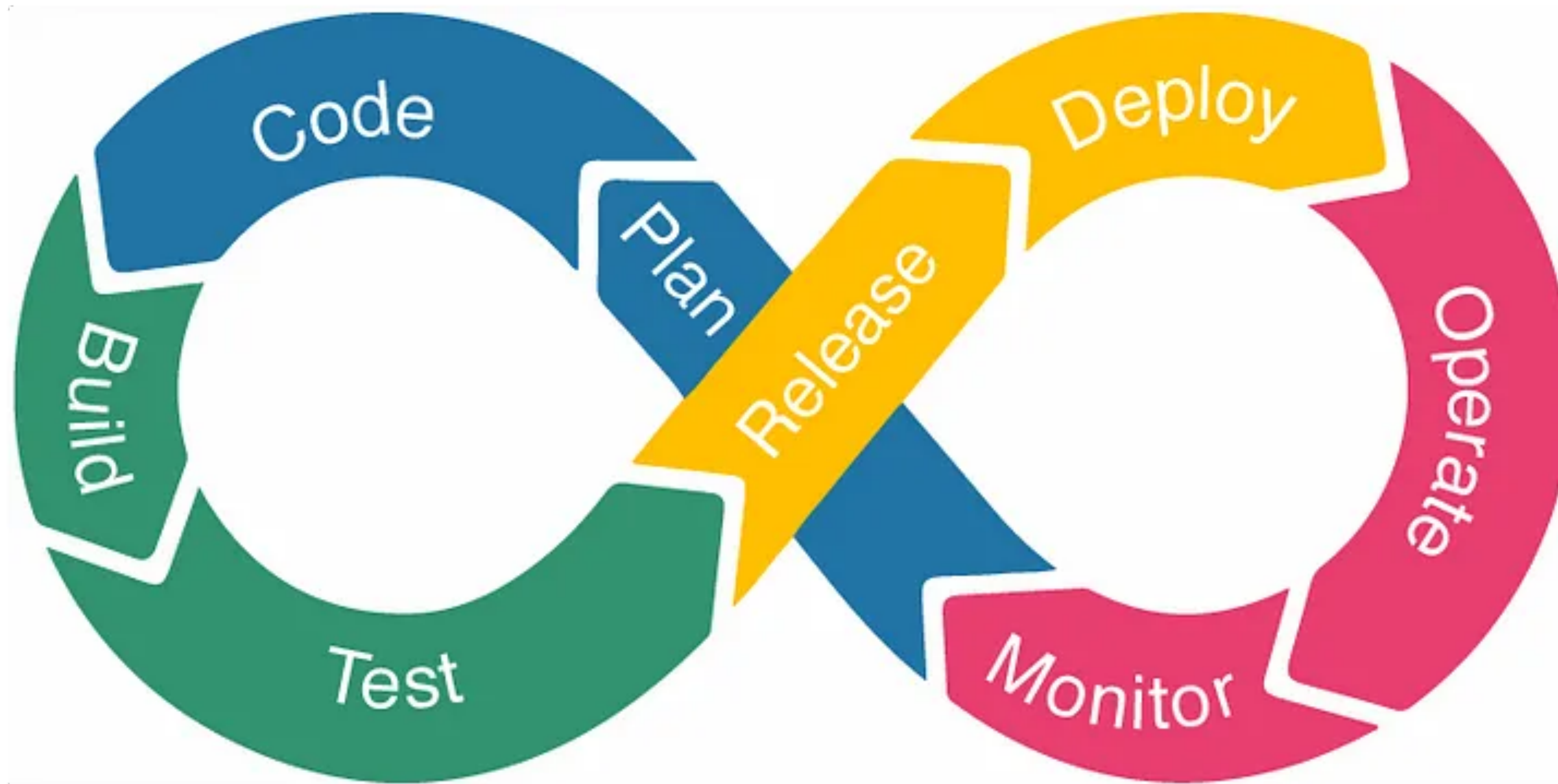


Image: <https://medium.com/türk-telekom-bulut-teknolojileri/devops-lifecycle-continuous-integration-and-development-e7851a9c059d>

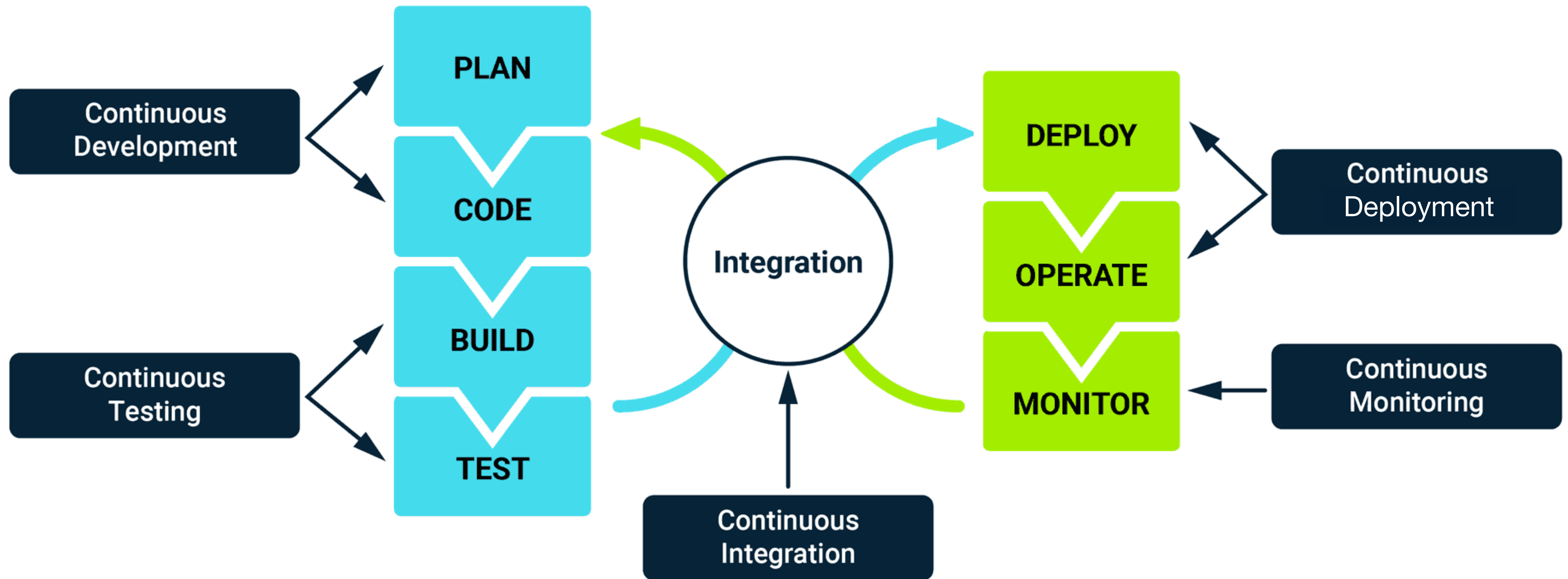


Image: <https://codilime.com/blog/devops-lifecycle/>

Collaboration

Communication

- Encourage collaboration between development, operations, and other stakeholders
- Foster a culture where all team members **share responsibility** for the entire process, from development to deployment
- Implement regular **feedback loops** to identify and address issues quickly

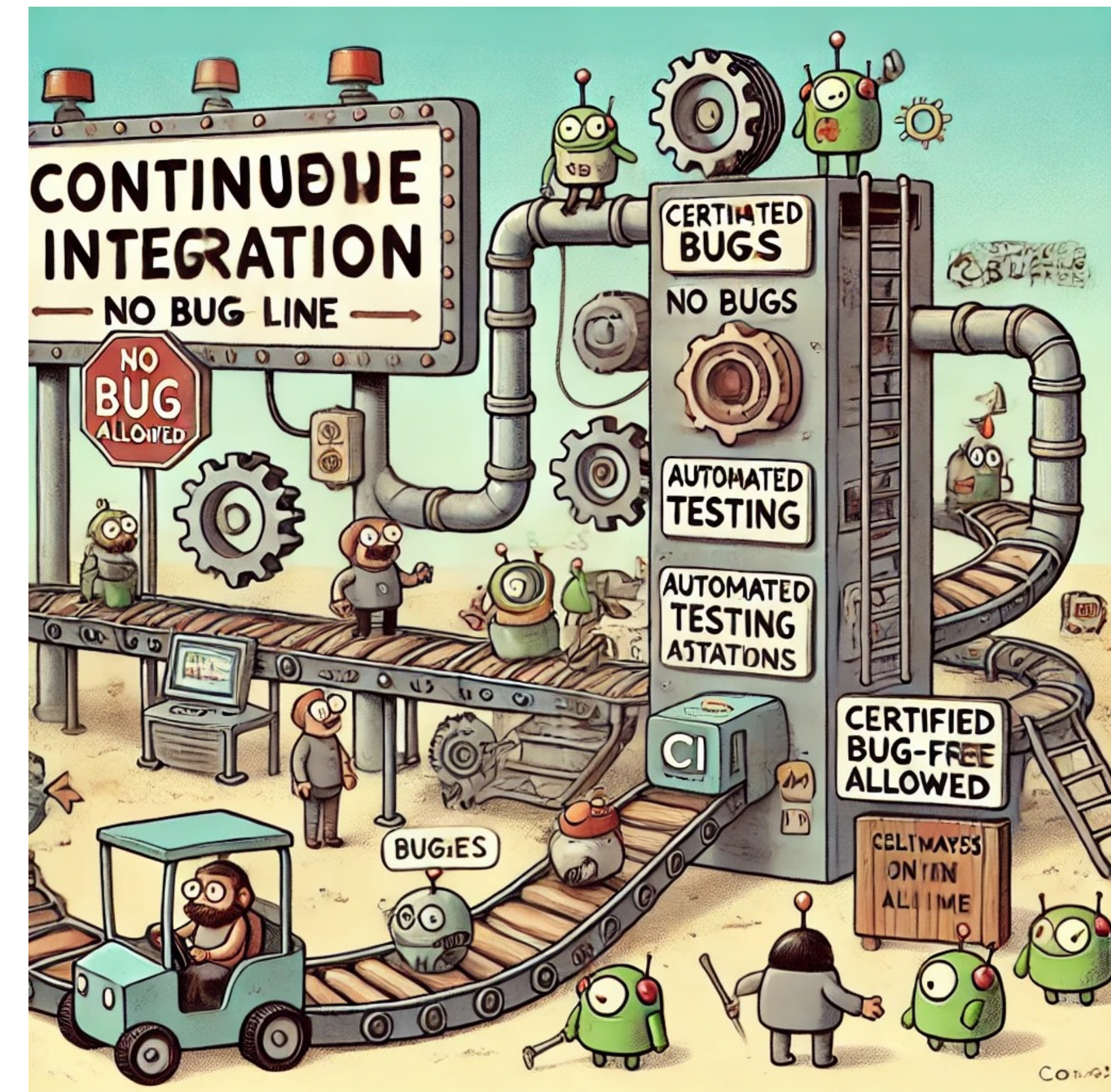
Communication

- **Blameless Postmortems**
 - **Post-incident reviews** focused on **learning and improvement**
 - Not assigning blame
- Tools and platforms to enhance collaboration and streamline workflows
 - GitHub issue tracker, Jira, Slack, Discord

Continuous Integration

CI is the first step in continuous delivery pipelines

- Code changes **frequently merged** into repository
- Each merge triggers an **automated build** process
- **Automated testing** catches bugs early
- Developers get **quick feedback** on their code
- Frequent, smaller integrations **reduce conflicts**



Continuous Delivery

Automated Deployment Pipelines

- Always ready for deployment to **production**
- **Automated** Pipeline Stages:
 - build, test, and deploy stages
- Regular, **reliable releases** with minimal manual intervention
- **Rollback** to previous versions if issues arise
- Deploy to **multiple environments** automatically



Monitoring and Observability

- Continuously **track system performance** and **health**
- Collect and analyze key **metrics, logs, and traces**
- Automated alerts for **early detection** of issues
- Continuous Improvement:
 - Use insights to **enhance system reliability** and **performance**



Traces

- Detailed record of the **entire flow of a request** through a system
- Help identify performance bottlenecks
- Understand the path taken by a request across distributed systems
- Understand how different services interact



Logs

- Detailed, time-stamped **records of events** and activities within a system
- Capture what happened at a specific time, such as errors, transactions, or user actions
- Important for debugging and troubleshooting



Logs — Structured Logging

- Log Attributes as key-value pairs
 - Save to JSONL files
 - Viewing logs: Loki Grafana plugin
- Libraries:
 - Go: slog (stdlib), zap (Uber), zerolog
 - Python: stdlib, Loguru, structlog
 - C#: Serilog, NLog, Microsoft.Extensions.Logging



JSONHandler (JSONL Format)

```
{
  "time": "2026-08-23T08:21:50.682241981+02:00",
  "level": "DEBUG",
  "source": {
    "function": "github.com/quickfeed/quickfeed/web/hooks.GitHubWebHook.handlePush",
    "file": "web/hooks/push.go",
    "line": 10
  },
  "msg": "push"
}, {
  "time": "2026-08-23T08:21:50.692557007+02:00",
  "level": "DEBUG",
  "source": {
    "function": "github.com/quickfeed/quickfeed/web/hooks.GitHubWebHook.handlePush",
    "file": "web/hooks/push.go",
    "line": 10
  },
  "msg": "push"
}, {
  "time": "2026-08-23T08:21:50.694366356+02:00",
  "level": "DEBUG",
  "source": {
    "function": "github.com/quickfeed/quickfeed/ci.(*RunData).RunTests",
    "file": "ci/run_tests.go",
    "line": 75
  },
  "msg": "close"
}, {
  "time": "2026-08-23T08:21:50.934868946+02:00",
  "level": "INFO",
  "source": {
    "function": "github.com/quickfeed/quickfeed/scm.(*appTokenManager).Token",
    "file": "scm/github_app.go",
    "line": 82
  },
  "msg": "token"
}, {
  "time": "2026-08-23T08:21:50.935233729+02:00",
  "level": "DEBUG",
  "source": {
    "function": "github.com/quickfeed/quickfeed/scm.(*GithubSCM).Clone",
    "file": "scm/fetcher.go",
    "line": 49
  },
  "msg": "clone"
}, {
  "time": "2026-08-23T08:21:51.631139418+02:00",
  "level": "DEBUG",
  "source": {
    "function": "github.com/quickfeed/quickfeed/scm.(*GithubSCM).Clone",
    "file": "scm/fetcher.go",
    "line": 62
  },
  "msg": "repo"
}
```



```
$ kelora -j 2026-08-23.jsonl -k time,level,msg,event_type,user,course_code,assignment -b
```

```
2026-08-23T18:09:17.779099687+02:00 DEBUG processing student push      push jdoe    DAT515
2026-08-23T18:09:17.779315641+02:00 ERROR failed to find assignment  push jdoe    DAT515 .github
2026-08-23T19:15:19.959754264+02:00 DEBUG resolved push course      push Meling  DAT515
2026-08-23T19:15:19.971945588+02:00 DEBUG processing student push      push Meling  DAT515
2026-08-23T19:15:19.97359364+02:00 DEBUG cloning submission repository push Meling  DAT515 2go
2026-08-23T19:15:20.201270822+02:00 INFO  fetched GitHub App access token push Meling  DAT515 2go
2026-08-23T19:15:20.201621423+02:00 DEBUG cloning repository          push Meling  DAT515 2go
2026-08-23T19:15:20.858207168+02:00 DEBUG repository clone directory   push Meling  DAT515 2go
2026-08-23T19:15:20.85835351+02:00 DEBUG cloned submission repository  push Meling  DAT515 2go
2026-08-23T19:15:20.861684375+02:00 DEBUG running assignment tests      push Meling  DAT515 2go
2026-08-23T19:15:20.95821806+02:00 INFO  created container            push Meling  DAT515 2go
2026-08-23T19:15:21.214090848+02:00 INFO  waiting for container         push Meling  DAT515 2go
2026-08-23T19:16:11.679871585+02:00 INFO  container exited             push Meling  DAT515 2go
2026-08-23T19:16:11.680175573+02:00 INFO  container completed           push Meling  DAT515 2go
2026-08-23T19:16:11.68748229+02:00 INFO  removing container            push Meling  DAT515 2go
2026-08-23T19:16:12.015473226+02:00 DEBUG test results extracted        push Meling  DAT515 2go
2026-08-23T19:16:12.274886368+02:00 DEBUG fetching previous submission  push Meling  DAT515 2go
2026-08-23T19:16:12.292535494+02:00 DEBUG updating submission           push Meling  DAT515 2go
2026-08-23T19:16:12.441374934+02:00 DEBUG recorded submission           push Meling  DAT515 2go
2026-08-23T19:16:12.442450166+02:00 DEBUG updated slip days             push Meling  DAT515 2go
```


Metrics

- Quantitative measures of the performance and behavior of a system over time
- CPU usage, memory consumption, and request rates
- Often used for monitoring and alerting
- Types: Time Guages and Counters



Metrics Systems

- Prometheus
- Grafana



Metrics Collection (in ci package)

```
// TestExecutionMetricsCollectors returns a list of Prometheus metrics collectors for test execution.
func TestExecutionMetricsCollectors() []prometheus.Collector {
    return []prometheus.Collector{
        cloneTimeGauge,
        validationTimeGauge,
        testExecutionTimeGauge,
        testsStartedCounter,
        testsFailedCounter,
        testsSucceededCounter,
    }
}
```


Metrics Collection (in metrics package)

```
var reg = prometheus.NewRegistry()

func init() {
    metricsCollectorsSets := [][]prometheus.Collector{
        interceptor.RPCMetricsCollectors(),
        ci.TestExecutionMetricsCollectors(),
    }
    for _, collectors := range metricsCollectorsSets {
        reg.MustRegister(collectors...)
    }
}

func Handler() http.Handler {
    return promhttp.HandlerFor(reg, promhttp.HandlerOpts{})
}
```


Metrics Collection (in ci package)

```
cloneTimeGauge = prometheus.NewGaugeVec(prometheus.GaugeOpts{
    Name: "quickfeed_clone_repositories_time",
    Help: "The time to clone tests and student repository for test execution.",
}, []string{"user", "course"})
```

```
validationTimeGauge = prometheus.NewGaugeVec(prometheus.GaugeOpts{
    Name: "quickfeed_repository_validation_time",
    Help: "The time to validate student repository for issues.",
}, []string{"user", "course"})
```

```
testExecutionTimeGauge = prometheus.NewGaugeVec(prometheus.GaugeOpts{
    Name: "quickfeed_test_execution_time",
    Help: "The time to run test execution.",
}, []string{"user", "course"})
```


Metrics Collection (in ci package)

```
func timer(user, course string, gauge *prometheus.GaugeVec) func() {
    responseTimer := prometheus.NewTimer(
        prometheus.ObserverFunc(
            gauge.WithLabelValues(user, course).Set,
        ),
    )
    return func() { responseTimer.ObserveDuration() }
}
```


Metrics Collection (in ci package)

```
// Start the timer and defer the duration observation until the function returns.
defer timer(user, course, testExecutionTimeGauge)()
out, err := runner.Run(ctx, job)
if err != nil && out == "" {
    testsFailedCounter.WithLabelValues(user, course).Inc()
    if errors.Is(err, ErrConflict) {
        return nil, err
    }
    return nil, fmt.Errorf("test execution failed without output: %w", err)
}
```


Query the Prometheus Metrics Endpoint

```
% curl 127.0.0.1:9097
```

```
# HELP quickfeed_test_execution_time The time to run test execution.
```

```
# TYPE quickfeed_test_execution_time gauge
```

```
quickfeed_test_execution_time{course="DAT515",user="a"} 51.153905822
```

```
quickfeed_test_execution_time{course="DAT515",user="b"} 23.1836886
```

```
quickfeed_test_execution_time{course="DAT515",user="c"} 22.458404788
```

```
quickfeed_test_execution_time{course="DAT515",user="d"} 22.125341367
```

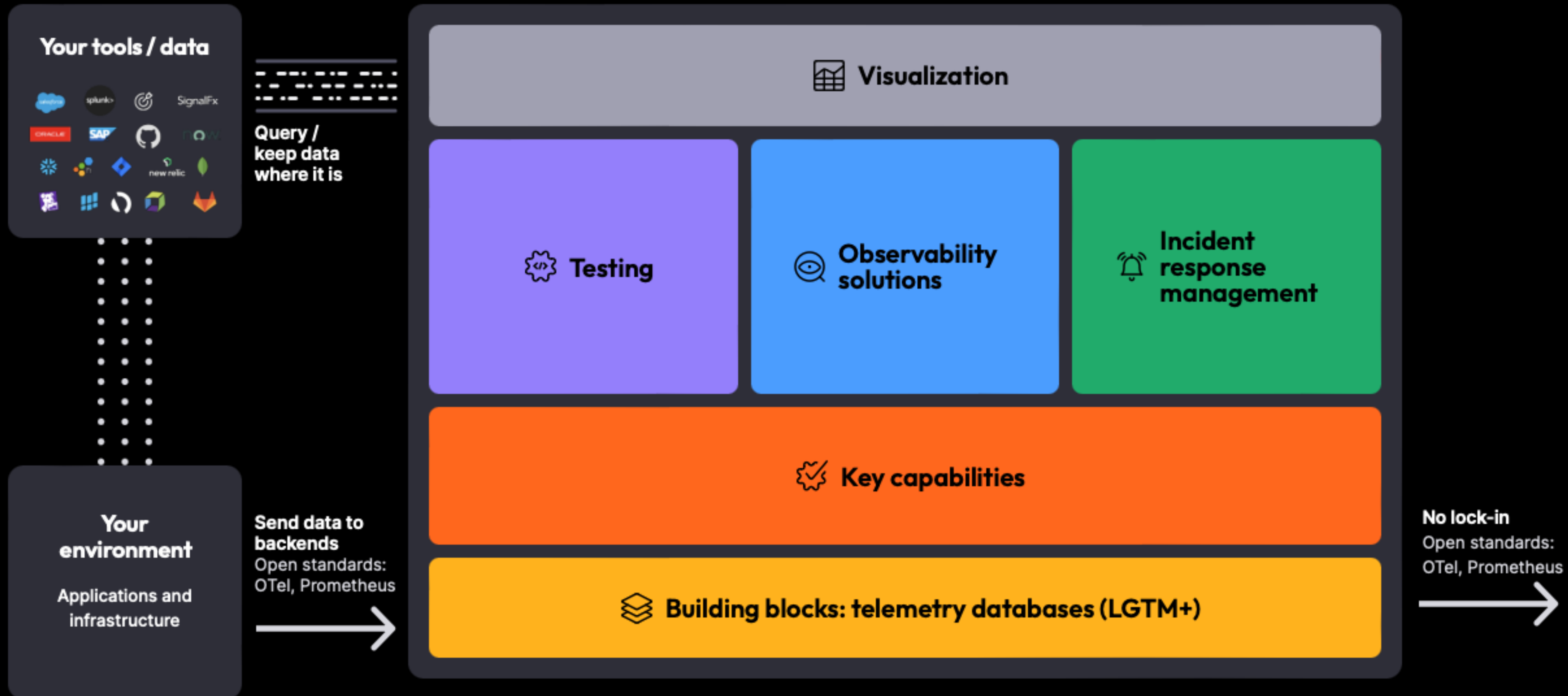
```
quickfeed_test_execution_time{course="DAT515",user="e"} 23.344935203
```

```
quickfeed_test_execution_time{course="DAT515",user="f"} 23.937536027
```

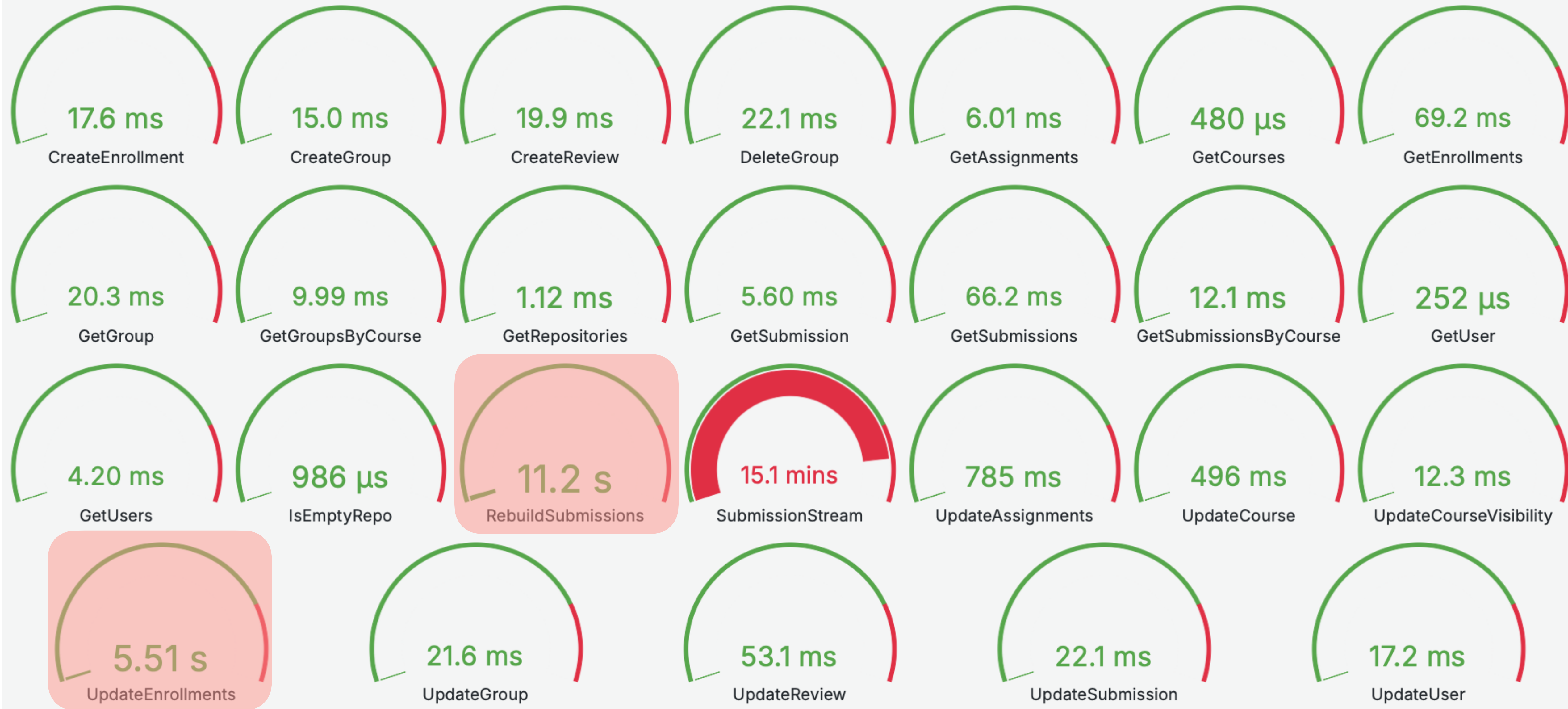
```
quickfeed_test_execution_time{course="DAT515",user="g"} 23.606847502
```

```
quickfeed_test_execution_time{course="DAT515",user="h"} 24.153919885
```

```
quickfeed_test_execution_time{course="DAT515",user="i"} 32.444826156
```

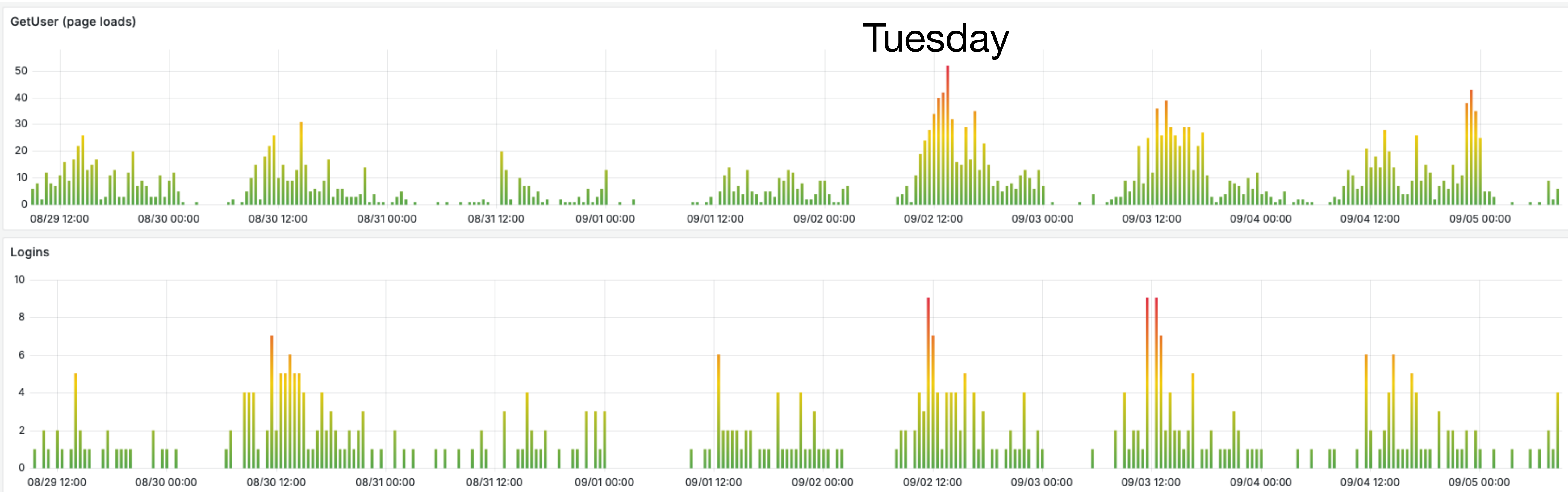
Response time (Last 30 days)



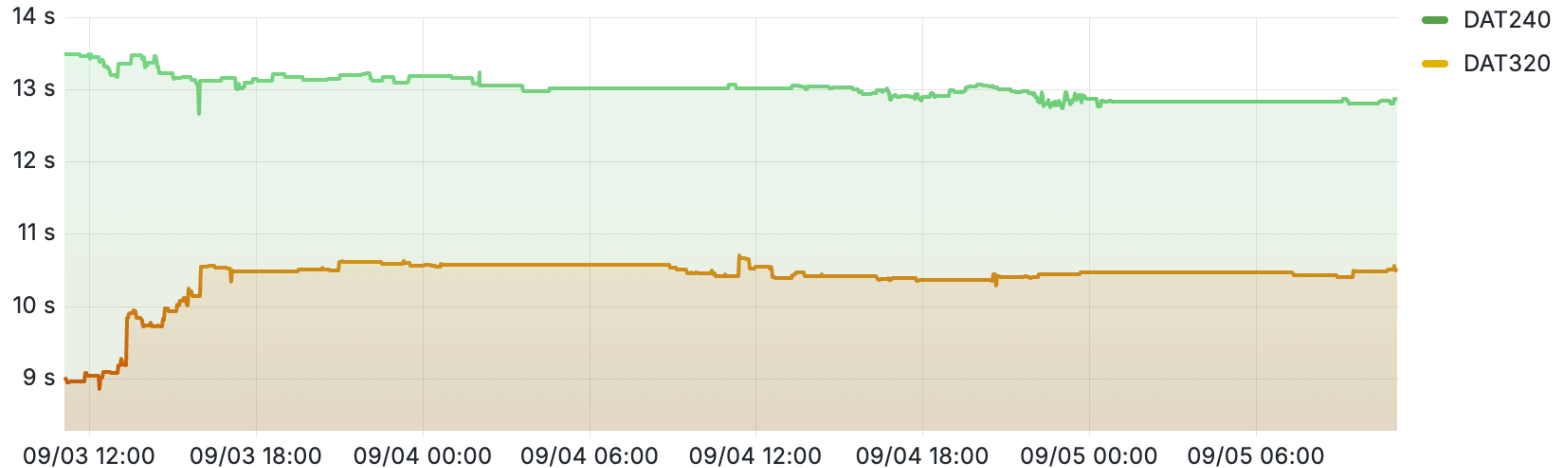
Grafana - Metrics Dashboard

Response time (Last 7 days)

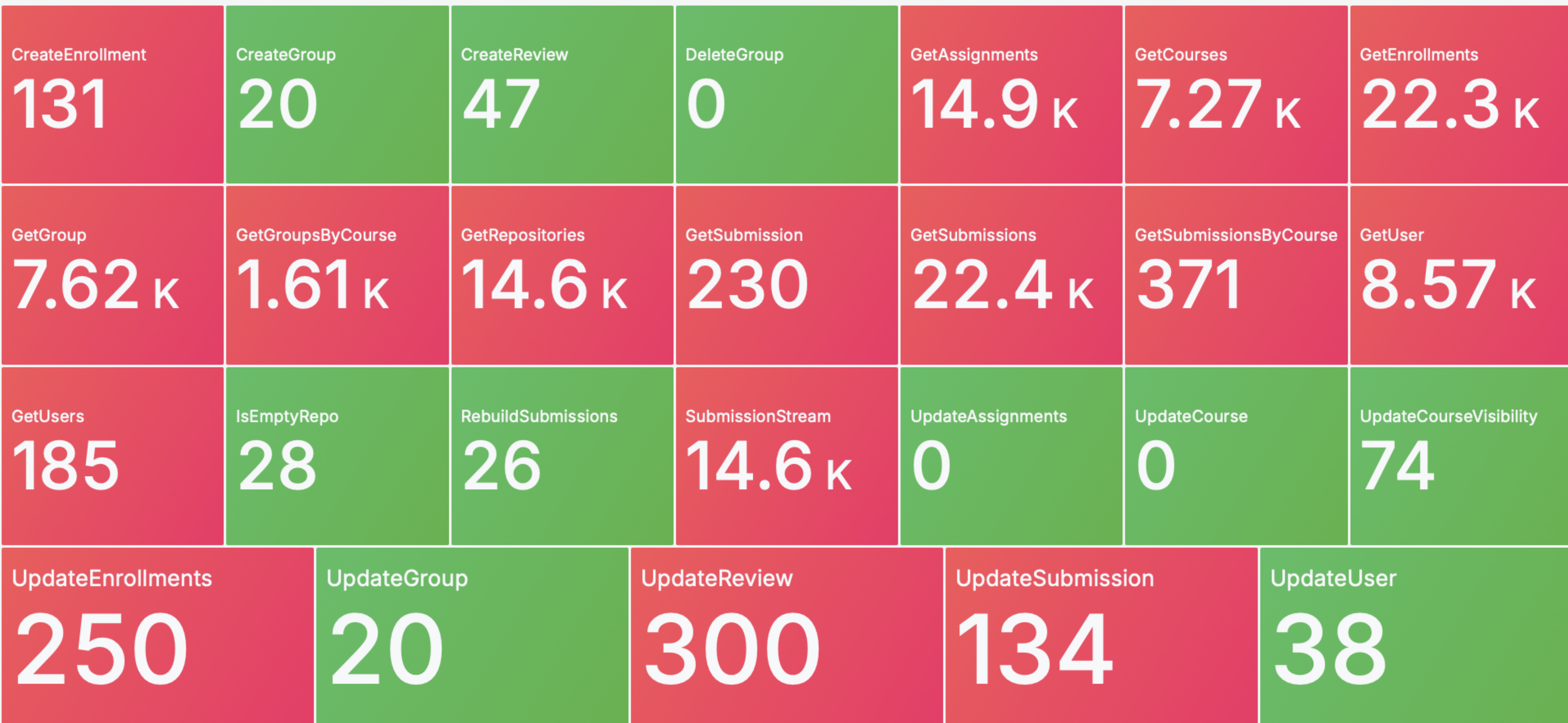




Test execution time (Last 2 days)



Method Accessed Count (Last 90 days)



Joke time

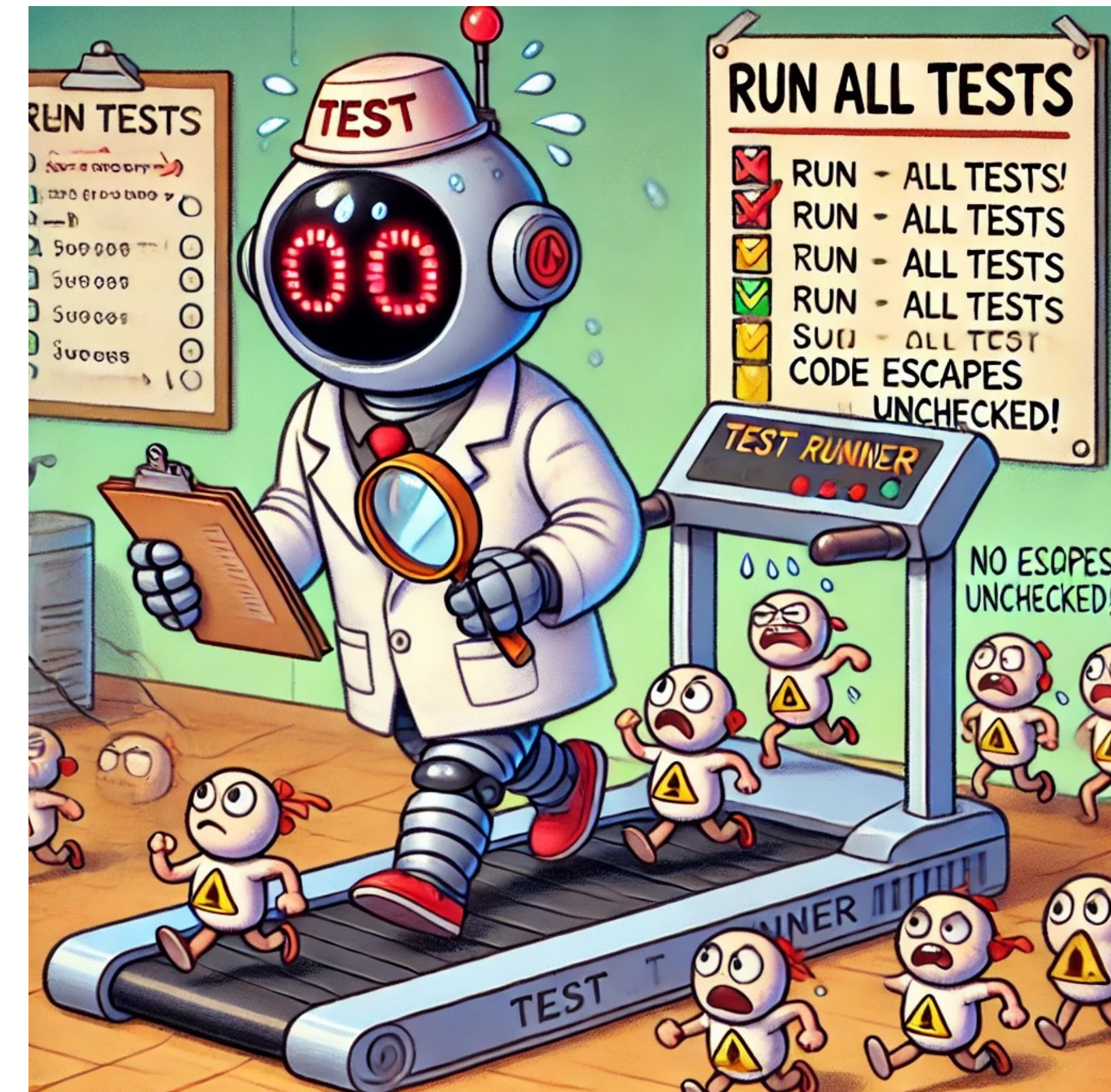
Why did the developer break
up with the DevOps engineer?

Too much automation,
not enough commitment!

Automated Testing

Continuous Testing at every stage of the development pipeline

- **Reduce risk** of human errors by automating repetitive test tasks
- Run tests automatically to **accelerate feedback**
- Quickly **catch regressions** with automated tests
- Several types of testing
 - Unit, integration, property-based, fuzz
 - Linters
- Test coverage



FABST-ON THE TESTS

TABLE-DRIVEN TEST

- | | |
|--|--|
| ☒ TEST CASES | ☒ TEST CASES |
| ☒ INPUTS | ☒ INPUTS |
| ☒ INPUTS | ☒ BUGS |
| ☒ TEST CASES | ☒ PASSED |
| ☒ PASSED | ☒ PASSED |
| ☒ PASSED | ☒ PASSED! |

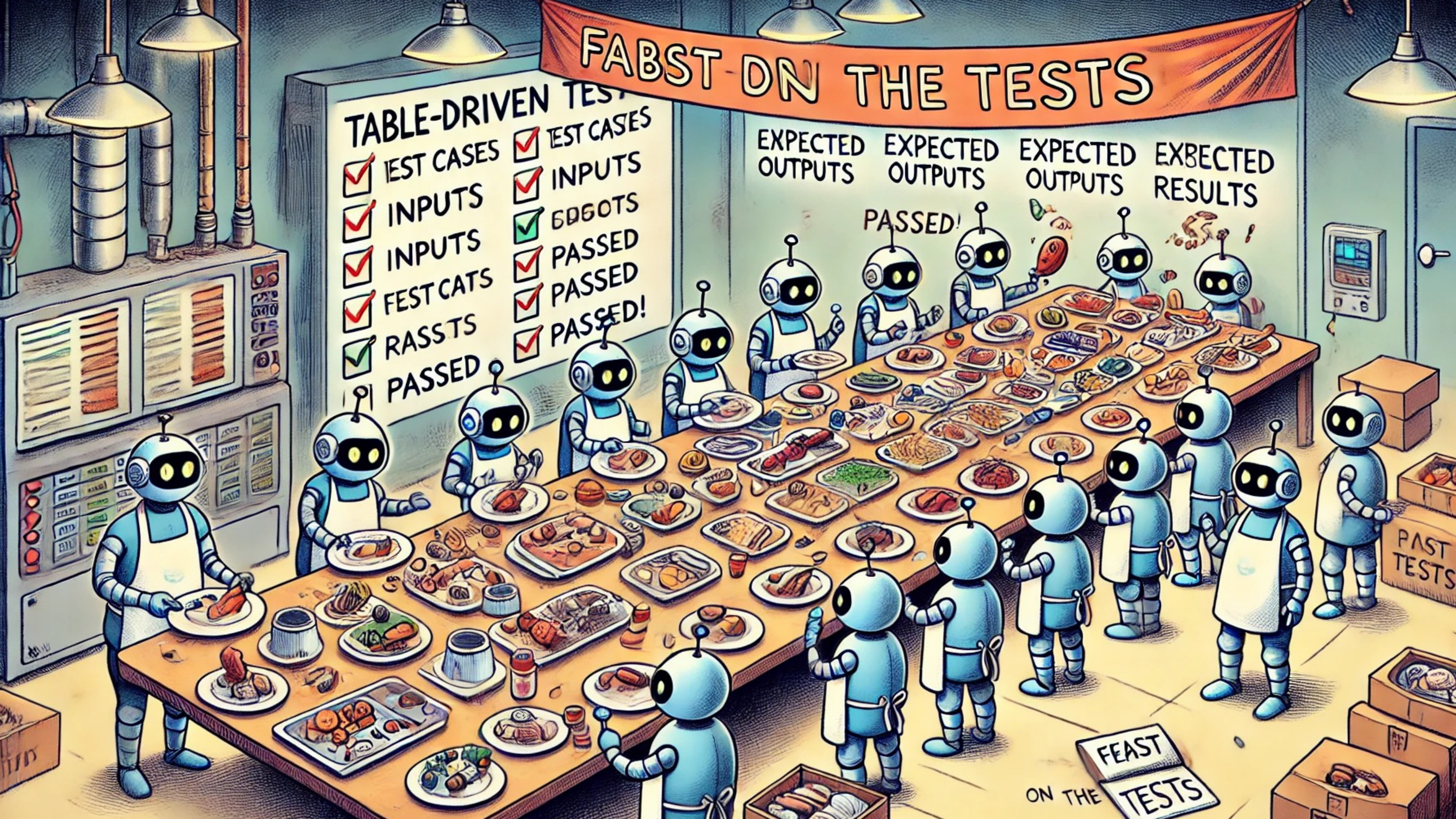
EXPECTED
OUTPUTS

EXPECTED
OUTPUTS

EXPECTED
OUTPUTS

EXPECTED
RESULTS

PASSED!




```
quickfeed > scm > go github_mock_test.go > ...
```

```
run test | debug test
```

```

189 func TestMockGetRepositories(t *testing.T) {
190     tests := []struct {
191         name    string
192         org     string
193         want    []*Repository
194         wantErr bool
195     }{
196         {name: "IncompleteRequest/NilOrg", org: "", want: nil, wantErr: true},
197         {name: "IncompleteRequest/NoOrgName", org: "", want: nil, wantErr: true},
198         {name: "CompleteRequest/NotFound", org: "buz", want: []*Repository{}, wantErr: false},
199         {name: "CompleteRequest/SixRepos", org: "foo", want: []*Repository{
200             {ID: 1, Repo: "info"},
201             {ID: 2, Repo: "assignments"},
202             {ID: 3, Repo: "tests"},
203             {ID: 4, Repo: "meling-labs"},
204             {ID: 5, Repo: "josie-labs"},
205             {ID: 6, Repo: "groupX"},
206         }},
207     }

```



```
208 s := NewMockedGithubSCMClient(qtest.Logger(t), WithOrgs(ghOrgFoo, ghOrgBar, ghOrgBuz), WithRepos(repos...))
209 for _, tt := range tests {
210     t.Run(tt.name, func(t *testing.T) {
211         got, err := s.GetRepositories(context.Background(), tt.org)
212         if (err != nil) != tt.wantErr {
213             t.Errorf("GetRepositories() error = %v, wantErr %v", err, tt.wantErr)
214             return
215         }
216         if diff := cmp.Diff(tt.want, got); diff != "" {
217             t.Errorf("GetRepositories() mismatch (-want +got):\n%s", diff)
218         }
219     })
220 }
221 }
```


Name function being tested with inputs and got, wanted output

```
s := NewMockedGithubSCMClient(qtest.Logger(t), WithOrgs(ghOrgFoo, ghOrgBar), WithRepos(repos...))
for _, tt := range tests {
    name := qtest.Name(tt.name, []string{"Owner", "Path"}, tt.opt.Owner, tt.opt.Repo)
    t.Run(name, func(t *testing.T) {
        if empty := s.RepositoryIsEmpty(context.Background(), tt.opt); empty != tt.wantEmpty {
            t.Errorf("RepositoryIsEmpty(%+v) = %t, want %t", *tt.opt, empty, tt.wantEmpty)
        }
    })
}
```



```
go test -v -run TestMockRepositoryIsEmpty
```

```
=== RUN    TestMockRepositoryIsEmpty
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest/Owner=foo
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=assignments
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=tests
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=meling-labs
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=assignments
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=tests
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=meling-labs
--- PASS: TestMockRepositoryIsEmpty (0.01s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest/Owner=foo (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=assignments (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=tests (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=meling-labs (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=assignments (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=tests (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=meling-labs (0.00s)
PASS
ok      github.com/quickfeed/quickfeed/scm      0.359s
```


Testing that tests fail on bad input

```
go test
--- FAIL: TestMockRepositoryIsEmpty (0.00s)
    --- FAIL: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=baz/Path=tests (0.00s)
        github_mock_test.go:248: RepositoryIsEmpty({ID:0 Repo:tests Owner:baz}) = true, want false
FAIL
exit status 1
FAIL    github.com/quickfeed/quickfeed/scm    8.606s
```


Testing that tests fail on bad input

```
go test -v -run TestMockRepositoryIsEmpty
=== RUN    TestMockRepositoryIsEmpty
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest/Owner=foo
=== RUN    TestMockRepositoryIsEmpty/IncompleteRequest/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=assignments
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=tests
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=meling-labs
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=info
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=assignments
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=baz/Path=tests
github_mock_test.go:248: RepositoryIsEmpty({ID:0 Repo:tests Owner:baz}) = true, want false
=== RUN    TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=meling-labs
--- FAIL: TestMockRepositoryIsEmpty (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest/Owner=foo (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/IncompleteRequest/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=assignments (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=tests (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/Empty/Owner=bar/Path=meling-labs (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=info (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=assignments (0.00s)
    --- FAIL: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=baz/Path=tests (0.00s)
    --- PASS: TestMockRepositoryIsEmpty/CompleteRequest/NonEmpty/Owner=foo/Path=meling-labs (0.00s)
FAIL
exit status 1
FAIL    github.com/quickfeed/quickfeed/scm    0.356s
```


feat: Add context logging support #1046

 Draft meling wants to merge 1 commit into `master` from `logging-with-context` 

 Conversation 0  Commits 1  Checks 8  Files changed 3



meling commented on Feb 7 • edited ▾

Member ...

This adds a simple interceptor-based context logging feature that can be used in RPC calls to fetch the context logger to log with session context such as source method, userID, courseID etc. We can add more global context if necessary, but most often we probably want to add logging information via the actual log msg.

This version just uses the current zap logger to avoid introducing another logging framework now. We can decide later whether or not to replace zap with slog.

Fixes [#670](#)

Fixes [#489](#)



  feat(interceptor): added context logging support via interceptor ...  ef5d465



 **Some checks were not successful**

[Hide all checks](#)

3 failing, 9 successful, and 1 cancelled checks

  Go Test / test (1.21, ubuntu-latest) (pull_request) Failing after 2m [Details](#)

  Test Coverage / test (1.21) (pull_request) Failing after 2m [Details](#)

  golangci-lint / lint (pull_request) Failing after 1m [Details](#)

  CodeQL / Analyze (go) (pull_request) Successful in 2m [Details](#)

QuickFeed Pull Request

- Create Pull Request with the gh tool
- `gh pr create --title "feat: Add cont..."`
- `gh pr view --web`

 master

[quickfeed](#) / [.github](#) / [workflows](#) / [go.yml](#) 

View Runs

 **meling** chore: updated dependencies and github workflows to go 1.22.5

41 lines (35 loc) · 916 Bytes

Code

Blame

```
1  name: Go Test
2
3  on:
4    push:
5      branches: [master]
6      paths:
7        - '**.go'
8    pull_request:
9      paths:
10       - '**.go'
11
12  jobs:
13    test:
14      strategy:
15        matrix:
16          go-version: ["1.22.5"]
17          platform: [ubuntu-latest, macos-latest]
18      runs-on: ${ matrix.platform }
19
```

- GitHub Actions Workflow


```
quickfeed > .github > workflows > go.yml
```

```

1  name: Go Test
2
3  on:
4    push:
5      branches: [master]
6      paths:
7        - '**.go'
8    pull_request:
9      paths:
10       - '**.go'
11
12  jobs:
13    test:
14      strategy:
15        matrix:
16          go-version: ["1.22.5"]
17          platform: [ubuntu-latest, macos-latest]
18      runs-on: ${ matrix.platform }
19
```



```

20  steps:
21    - name: Cache
22      uses: actions/cache@v4
23      with:
24        path: |
25          ~/.cache/go-build          # ubuntu-latest
26          ~/Library/Caches/go-build # macos-latest
27          ~/go/pkg/mod
28        key: ${{ runner.os }}-go-${{ hashFiles('**/go.sum') }}
29        restore-keys: |
30          ${{ runner.os }}-go-
31
32    - name: Checkout code
33      uses: actions/checkout@v4
34
35    - name: Set up Go
36      uses: actions/setup-go@v5
37      with:
38        go-version: ${{ matrix.go-version }}
39
40    - name: Run Go tests
41      run: DOCKER_TESTS=1 go test -v github.com/quickfeed/quickfeed/...

```


Actions

New workflow

All workflows

CodeQL

Dependabot Updates

Go Test

golangci-lint

Jest Test

Test Coverage

Management

Caches

Attestations

Runners

Go Test

[go.yml](#)

Filter workflow runs

...

321 workflow runs

Event ▾ Status ▾ Branch ▾ Actor ▾



Merge pull request #1124 from quickfee...

master



last month

...



1m 48s

Go Test #2224: Commit [fd2e4a7](#) pushed by meling



Ignore push on group repo's non-defaul...

bugfix/issue1053



last month

...



1m 52s

Go Test #2223: Pull request [#1124](#) opened by meling



Merge pull request #1122 from quickfee...

master



last month

...



1m 58s

Go Test #2222: Commit [1031154](#) pushed by meling



fix(ci): handle image conflict during bu...

fix-docker-conflict-error



last month

...



2m 5s

Go Test #2221: Pull request [#1122](#) opened by JosteinLindhom





Some checks were not successful
 7 successful and 1 failing checks

[Hide all checks](#)

✗	 DeepSource: JavaScript — Analysis failed: Blocking issues or failing ...	Details
✓	 Code scanning results / CodeQL Successful in 2s — No new alerts ...	Details
✓	 DeepSource: Docker — Analysis passed: No blocking issues or failin...	Details
✓	 DeepSource: Go — Analysis passed: No blocking issues or failing me...	Details
✓	 DeepSource: Secrets — Analysis passed: No blocking issues or failin...	Details
✓	 DeepSource: Shell — Analysis passed: No blocking issues or failing ...	Details

↗ Performance issues

121

{ } Style issues

3

📄 Documentation issues

0

⚠ Anti patterns

158

🛡 Security issues

0

☂ Coverage issues

0

ACTIVE ISSUES

305

ISSUES PREVENTED

4.6k

DEFAULT BRANCH

🔗 master

ANALYZERS

☂ JS > ⚙ +2

✓ Last analyzed [6d374c6](#) 3 days ago

OWASP® Top 10 ?

...

● PASSING

📉 0 active issues

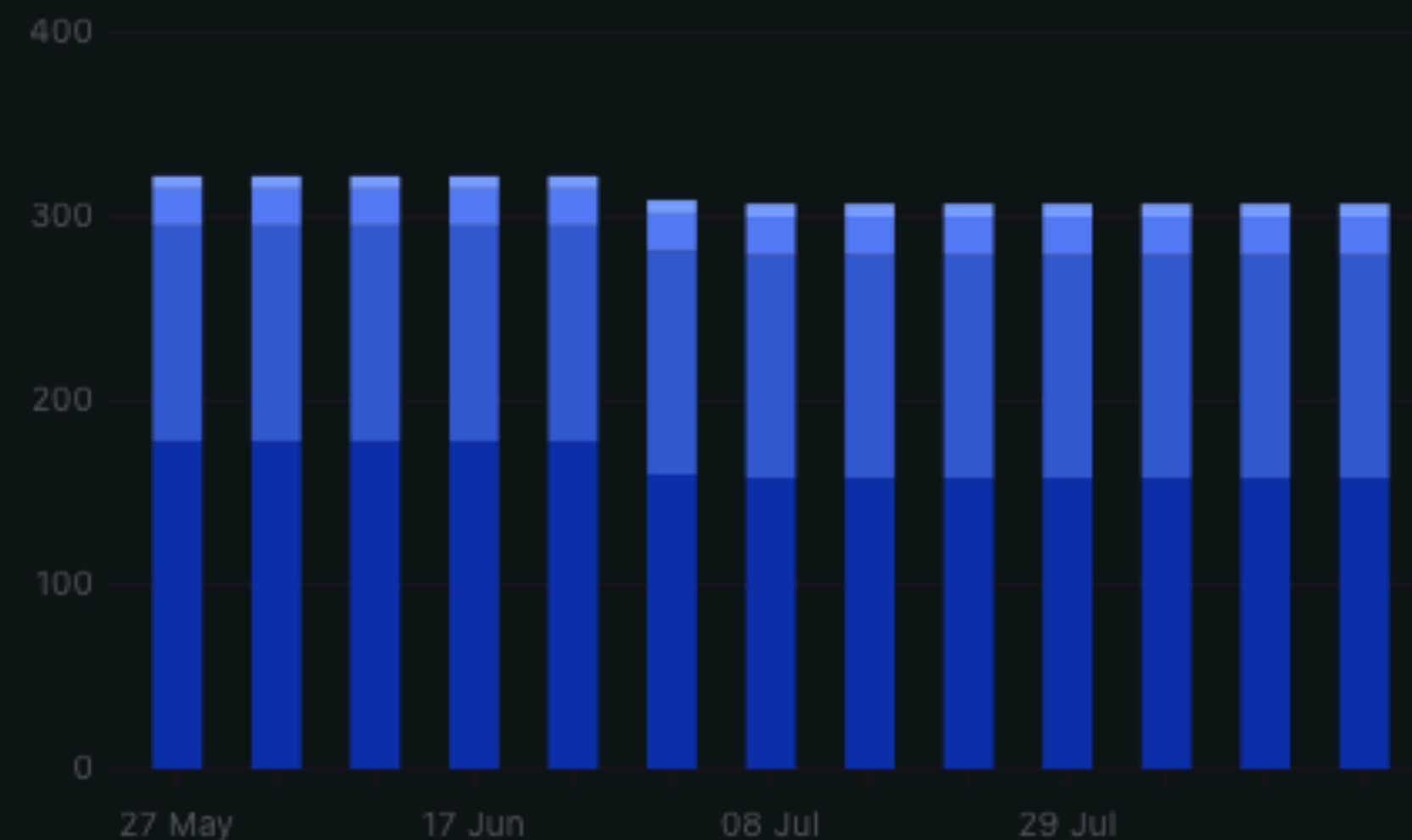


Issue Distribution by Category ?

...

📉 305 active issues

■ Anti-pattern ■ Performance
■ Bug risk ■ Others



☰ All issues	305
🌟 Recommended	114
🔒 Secrets	2
🚨 Bug Risk	21
⚠️ Anti-pattern	158
🛡️ Security	0
📈 Performance	121
🔍 Typecheck	0
☔ Coverage	0
{ } Style	3
📄 Documentation	0

Avoid `.bind()` or local functions in JSX properties JS-0417

📈 Performance 📊 Major ⌚ 2 months ago — 3 years old

📄 Seen in 44 files

103

Non-interactive elements assigned mouse/keyboard event listeners JS-0760

📈 Performance 📊 Minor ⌚ 9 months ago — 3 years old

📄 Seen in 4 files

4

Prefer not to use words image, photo in image alt content JS-0750

📈 Performance 📊 Minor ⌚ 9 months ago — 3 years old

📄 Seen in 1 file

5

Non-interactive elements should not be assigned interactive roles JS-0761

📈 Performance 📊 Minor ⌚ 9 months ago — 3 years old

📄 Seen in 1 file

1

Joke time

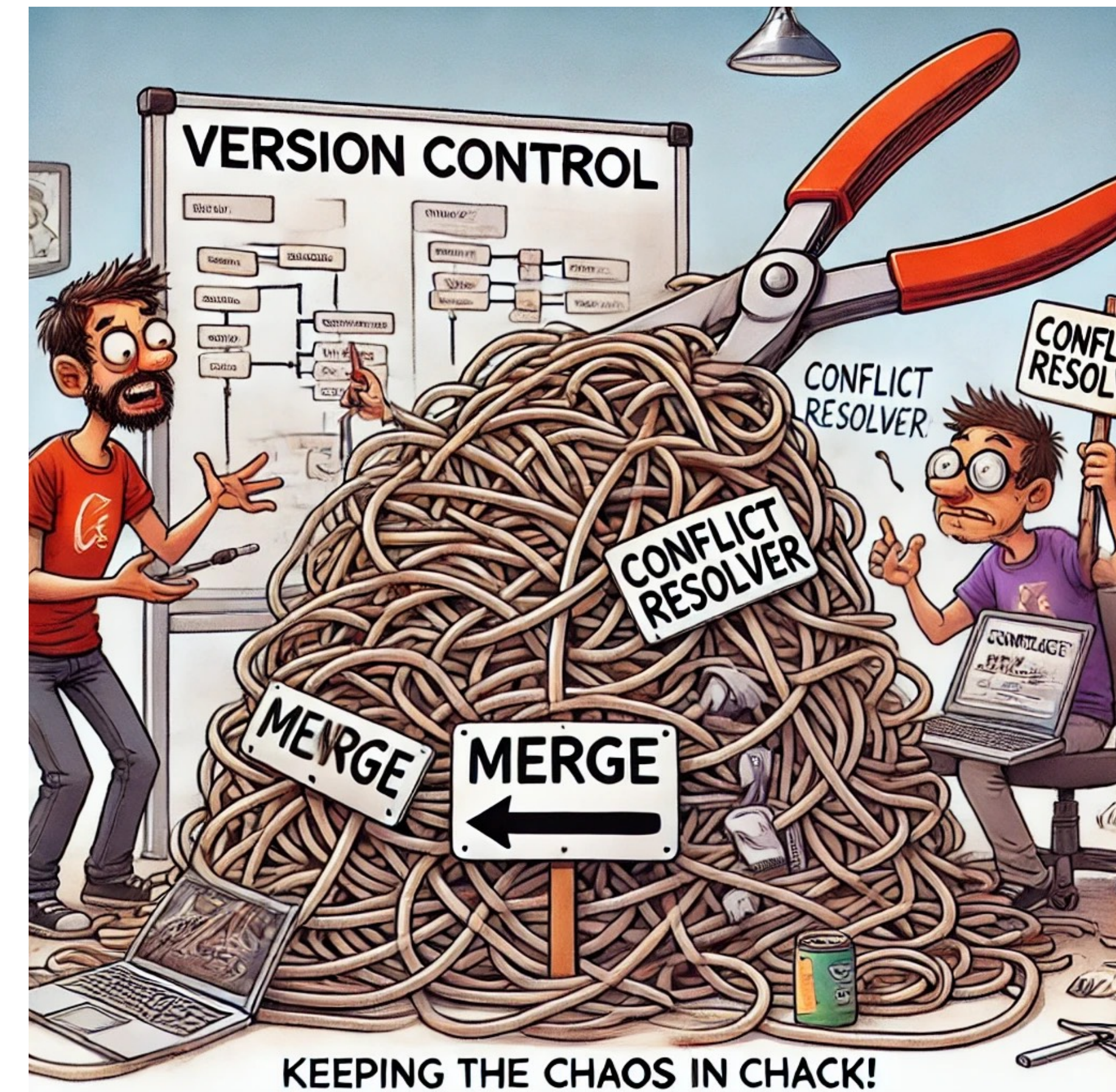
How do DevOps engineers
stay calm?

They automate their
stress away.

Version Control

Source Code Management

- Use branches for **feature** development, bug **fixes**, and **releases**
- Collaboration with shared repositories and **pull requests**
- Maintain a detailed **history of changes**
 - Who made them and why
 - Conventional commits
- **Revert to previous versions** if issues arise



Jujutsu — headline features

- Works seamlessly with Git repositories and workflows
- Branches themselves are tracked and versioned, not just commits
- Safe, local-first design – experimentation without risk of losing work
- Powerful undo and redo – Easily backtrack or reapply changes
- Flexible and intuitive – Command set designed to be simpler and more consistent than Git

DevSecOps

DevSecOps: Embed security practices into every phase

- Shift Left: Move security checks earlier to **catch vulnerabilities sooner**
- Tools to automate **security and vulnerability scans**
 - Govulncheck, Dependabot, Snyk, Aqua Security
- Continuously **monitor** applications and infrastructure for **security threats**
- Define **security policies** and controls as code, ensuring **consistent enforcement**



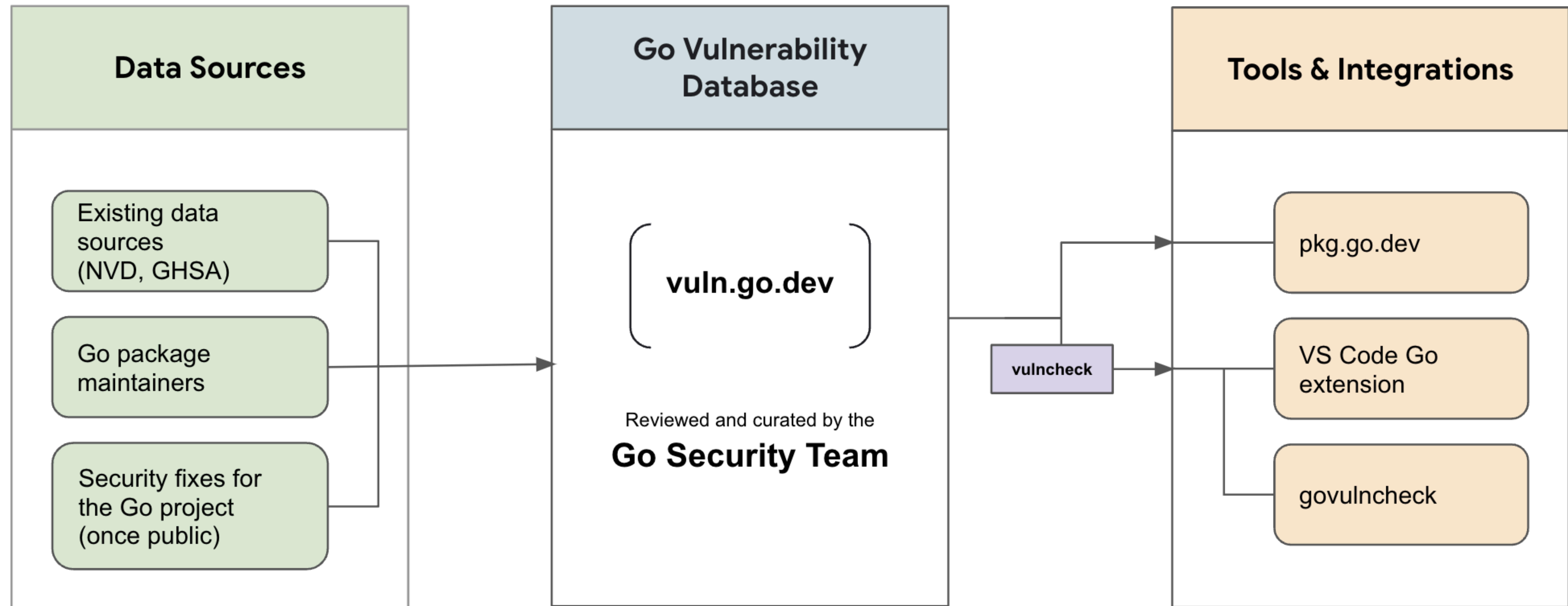
 Overview

Security overview

- Reporting
-  Policy
-  Advisories

- Vulnerability alerts
-  Dependabot
-  Code scanning
-  Secret scanning

Security policy • Disabled Define how users should report security vulnerabilities for this repository Set up a security policy
Security advisories • Enabled View or disclose security advisories for this repository View security advisories
Private vulnerability reporting • Enabled Allow users to privately report potential security vulnerabilities See reported vulnerabilities
Dependabot alerts • Enabled Get notified when one of your dependencies has a vulnerability View Dependabot alerts
Code scanning alerts • Enabled Automatically detect common vulnerability and coding errors View alerts
Secret scanning alerts • Disabled Get notified when a secret is pushed to this repository Enable in settings



- `cd work/quickfeed`
- `go install golang.org/x/vuln/cmd/govulncheck@latest`
- `govulncheck ./...`
 - Wait a minute or so...
 - No vulnerabilities found.

```
~/work/gorums git:(master)±2 (13.372s)
govulncheck ./...

=== Symbol Results ===

Vulnerability #1: GO-2024-2687
  HTTP/2 CONTINUATION flood in net/http
  More info: https://pkg.go.dev/vuln/GO-2024-2687
  Module: golang.org/x/net
  Found in: golang.org/x/net@v0.22.0
  Fixed in: golang.org/x/net@v0.23.0
  Example traces found:
  #1: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.ConnectionError.Error
  #2: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.ErrCode.String
  #3: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.FrameHeader.String
  #4: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.FrameType.String
  #5: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.ReadFrame
  #6: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteContinuation
  #7: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteData
  #8: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteGoAway
  #9: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteHeader
  #10: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WritePing
  #11: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteRSTStream
  #12: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteSettings
  #13: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteSettingsAck
  #14: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.Framer.WriteWindowUpdate
  #15: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.Setting.String
  #16: node.go:162:21: gorums.RawNode.FullString calls fmt.Sprintf, which eventually calls http2.SettingID.String
  #17: server.go:170:27: gorums.Server.Serve calls grpc.Server.Serve, which eventually calls http2.SettingsFrame.ForeachSetting
```


Vulnerability Database > All Reports > GO-2024-2687

Vulnerability Report: GO-2024-2687 standard library

Affects: net/http, golang.org/x/net | Published: Apr 03, 2024 | Modified: May 20, 2024

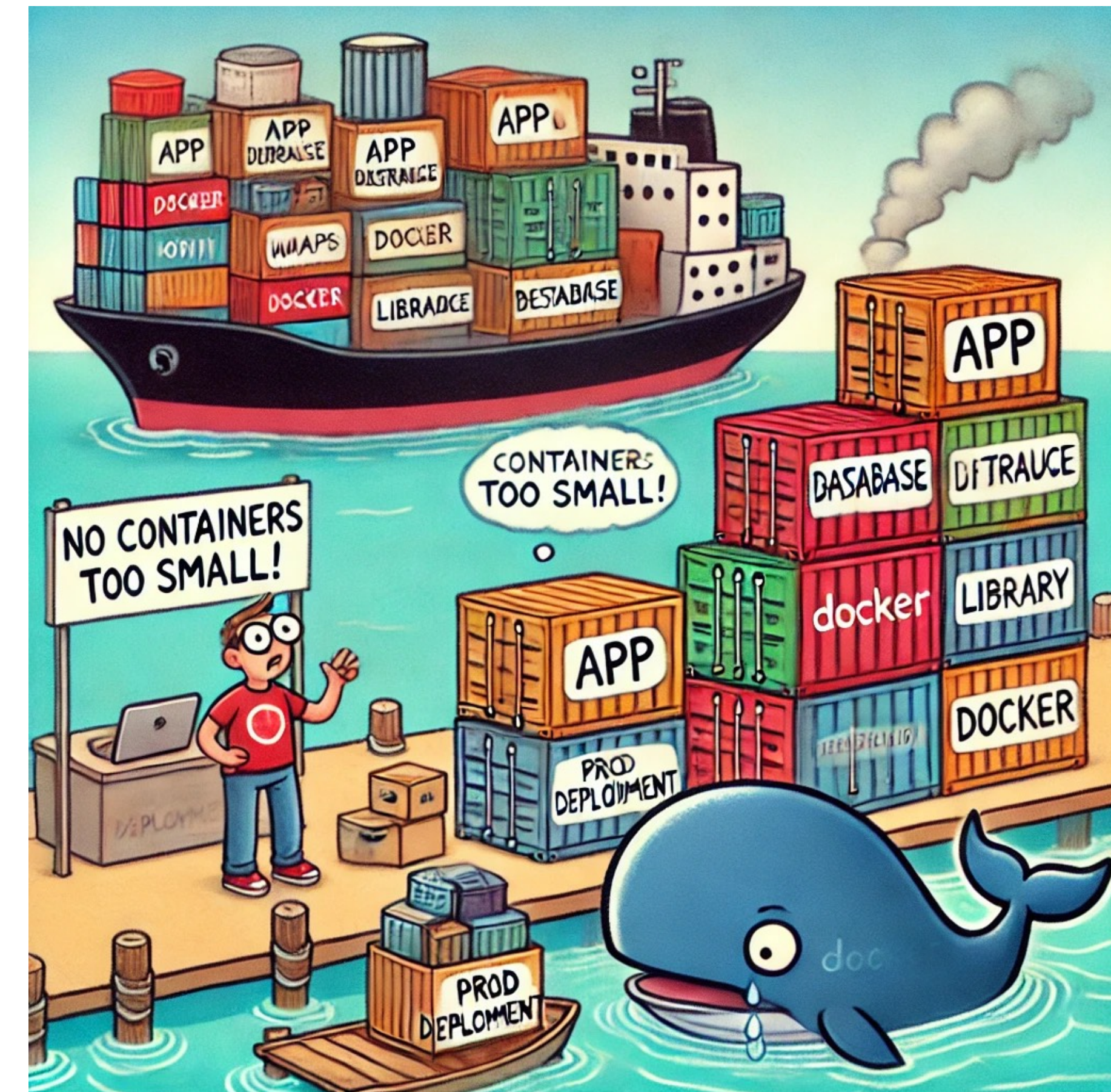
An attacker may cause an HTTP/2 endpoint to read arbitrary amounts of header data by sending an excessive number of CONTINUATION frames. Maintaining HPACK state requires parsing and processing all HEADERS and CONTINUATION frames on a connection. When a request's headers exceed MaxHeaderBytes, no memory is allocated to store the excess headers, but they are still parsed. This permits an attacker to cause an HTTP/2 endpoint to read arbitrary amounts of header data, all associated with a request which is going to be rejected. These headers can include Huffman-encoded data which is significantly more expensive for the receiver to decode than for an attacker to send. The fix sets a limit on the amount of excess header frames we will process before closing a connection.

Affected Packages

Path	Go Versions	Symbols
net/http	before go1.21.9, from go1.22.0-0 before go1.22.2	► 71 affected symbols
golang.org/x/net/http2	before v0.23.0	► 37 affected symbols

Containerization

- Package applications and dependencies together, creating isolated environments
- Run containers consistently across (development, testing, production)
- Start, stop, and scale applications quickly with minimal overhead
- Uniform behavior of applications across multiple environments
- Docker: Widely adopted in DevOps workflows



Orchestration

Automates deployment, scaling, and management of containerized applications

- **Load Balancing:** Distributes traffic across containers to ensure optimal performance and availability
- **Self-Healing:** Automatically restarts failed containers
- **Scaling:** Dynamically scale applications up or down based on demand
- **Service Discovery:** Automatically connects services within a cluster
- **Kubernetes:** Widely adopted in DevOps workflows



Best Practices Summary

Summary

- Encourage small, **frequent commits** to reduce integration challenges
- **Automate Everything**: Build, test, and deployment processes to ensure consistency
- Rapid **feedback** on code changes to identify issues early
- **Comprehensive testing**, including unit, integration, and end-to-end tests, linters and security vulnerability checkers (DevSecOps)
- **Separate environments** for development, testing, and production to avoid conflicts
- Continuously **monitor** pipeline performance and **optimize** for speed and reliability

Questions?

- <https://www.ibm.com/topics/devops>
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